Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
PI Name:	Perera, Jeevan Ph.D., J.D.		
Project Title:	NextGen Crew Countermeasure Software for Exploration Mission Support		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jeevan.s.perera@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	713-444-9136
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 NASA Parkway, #1		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	06/01/2016	**End Date:**	05/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	English, Kirk Ph.D. (Wyle Laboratories, Inc.) Hanson, Andrea Ph.D. (NASA Johnson Space Center) Kandah, Ruby Ph.D. (Wyle Science and Technology Group) Amonette, William Ph.D. (Science PI: University of Houston)		
Grant/Contract No.:	NCC 9-58-MA04501		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Each exercise device on the International Space Station (ISS) has a unique, customized software system interface with limited functionality, unique layouts / hierarchy, and operational principles that require significant crew training. The goals of this project were to: 1) develop a new, device-agnostic, one-portal software system (NextGen) capable of easily integrating external devices and sensors; 2) develop a new virtual exercise training software system (VETSS) and test its effectiveness to inform and correct resistance exercise technique; and 3) test the acute and chronic effects of exercise with a virtual exercise software (VES) on exercise performance and enjoyment. Specific Aim 1: To accomplish the first objective, the NextGen software was developed and compared to the existing Advanced Resistive Exercise Device (ARED) software. Efficiency, learnability, memorability, and overall usability were compared in 15 subjects over two sessions separated by two weeks. Results indicate that participants took 21% less time to complete exercise sessions in the NextGen UI (user interface) condition (7.6 ± 2.7 min) than in the ARED UI condition (9.6 ± 2.8 min). Additionally, measures of learnability show that the design of the NextGen UI better assisted participants in learning the system, evidenced by a 30% decrease in task completion time between Trials for the NextGen UI (Trial 1 = 9.0 ± 2.8 min; Trial 2 = 6.2 ± 1.8 min), whereas the ARED UI showed at 25% decrease in trial completion time (Trial 1 = 10.9 ± 2.8 min; Trial 2 = 8.2 ± 2.1 min). Finally, when looking across the two sessions to determine memorability only the ARED UI showed an increase in task completion time between the end of session 1 and the beginning of session 2. Specific Aim 2: The second objective was accomplished by developing the VETSS, using the Microsoft Kinect depth-based camera system. Basic software architecture was developed before collecting deadlift exercise input data on 20 participants. Integrating feedback from experienced strength and conditioning coach, a machine learning algorithm was used to pretrain the VETSS to identify five common deadlift errors. Subsequently, 16 new participants completed 4 sessions separated by 1-2 weeks. The first session was used to instruct subjects in proper deadlift technique; this was followed a week later by a one-rep maximum (1RM) determination and collection of a well-performed reference deadlift with a 3D motion capture system. Participants were randomized to either a control group (CON) with no feedback or a group that utilized VETSS (EXP). Reduction peak spine angle (SAp) is critical for injury reduction during the deadlift. CON experienced greater SAp and greater deviations from the reference SAp compared to EXP. From these data we conclude that exercise using the VETSS may improve hip kinematics and reduce spinal flexion; however, it does not seem to improve knee flexion kinematics. Deadlift exercise with the VETSS may provide corrective feedback, helping to prevent more severe spinal flexion and improving the consistency of hip kinematics. Specific Aim 3: The third objective of this project was to test the integration of an existing virtual environment software (VES) tool for its effectiveness to improve performance in a single bout of high intensity treadmill exercise, and adherence and performance during a one-month metabolic conditioning program. Two randomized controlled trials were completed to accomplish this objective. First, a randomized cross-over study design was employed (N=23). After determining their peak oxygen consumption (VO2peak), subjects completed three timed run-to-failure (TRF) tests at the same high intensity. TRF tests were conducted with the following conditions in random order: 1) exercise with no external stimuli (CON), 2) exercise with subject-selected music (MUS), or 3) exercise with subject-selected music and VES. During the TRF, test time (s), rating of perceived exertion (RPE), heart rate (HR; bpm), oxygen consumption (VO2; mL/kg/min), and respiratory exchange ratio (RER) were collected. The VES condition allowed users to select from 18 pre-recorded environments from scenic locations around the world and was displayed on a 121 cm screen in front of the participant. The results indicate that participants using VES ran for significantly longer than CON (12.6 ± 3.8%), but MUS was not significantly different than VES (8.4 ± 3.6%) or CON (5.6 ± 4.3%). HR was elevated in VES and MUS compared to CON. No other metabolic or perceptual differences were evident. Next participants who were physically active but not runners (N=18) completed the second phase of the project. They were matched for VO2 and BMI (body mass index) and randomly assigned to a self-selected music (MUS) or with music and VES group. Subjects completed a week of testing that consisted of TRF trials. Subsequently, they completed five or three days of treadmill exercise per week for four weeks (12-20 sessions); workout durations ranged from 10 – 30 min. Participants in both groups completed a similar number of intervals and the average peak HR was similar between groups in training. Peak RER during VO2peak tests were similar between groups at baseline and did not change during the post test. VO2peak increased in VES post-training (7.3±3.1%) but there was no change in MUS (3.9±2.2%). Peak ventilation also increased in both VES and MUS but there was no between group difference. HR during the post-test significantly decreased in VES. Total test time during the weekly TRF increased in weeks 1, 2, 3, 4, and post-training compared to baseline for VES group but was unchanged for MUS. Percent change in TRF for VES and CON groups was 96.6±33.5% and 34.1±13.4%, respectively. There was no change in positive engagement, revitalization, or tranquility, but perceived physical exhaustion was increased after the first week compared to baseline in both groups.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NextGen One-Portal UI will enhance the next generation software interface, allowing for improved user experience. The results of the usability evaluation indicate that that iterative design approach used in this effort resulted in a system that will meet the needs of crewmembers performing resistive and metabolic conditioning exercise. The virtual exercise trainer software system (VETSS) has potential to enhance the safety of exercise for individuals in fitness or therapy settings. Resistance exercise is effective at increasing whole body muscular strength, power, and bone mineral density. However, if performed incorrectly, it can result in serious and debilitating musculoskeletal injuries. This tool, when fully developed, could be used in commercial, medical, and home settings to mitigate injury risk. Research and development will continue to expand the scope and usability of the software. For example, the current software has only been tested using the deadlift exercise. To improve the impact, the exercise repository needs to be populated with additional exercises commonly used in a strength and conditioning program. Likewise, the virtual running environment is commercially available for home use. The program was modified to include a virtual training partner; our data analyses support the use of the software to improve exercise performance and ultimately aerobic fitness.

Task Progress:	The goal of Aim 1 was to streamline and enhance the capabilities of in-flight exercise software by 1) standardizing the software architecture for all exploration class exercise equipment, 2) modularizing the software for simplified integration with future and existing equipment, and 3) including plug-ins connectivity for access to commercially available virtual fitness environments and analysis/feedback/instructional capabilities such as the Microsoft Kinect. Multiple experiments were conducted to assess the efficacy of the NextGen One-Portal UI for exploration class exercise equipment. A human factors approach was used to evaluate the usability of the newly developed NextGen software compared to the current ARED software through subjective and objective measures which quantified the efficiency (e.g., time to complete steps and tasks), effectiveness (e.g., time required for software proficiency, number of errors), learnability (e.g., as measured how quickly subject improved), memorability (e.g., measured by retention of UI components), and user satisfaction (System Usability Scale and User Experience Questionnaire). Although all assessments and testing have been completed as planned, there are areas of improvement that can be explored in the near term. Significant progress was also made during the past funded year in developing virtual exercise training software system (VETSS) using the Microsoft Kinect. Specifically, the software architecture was developed, refined, and the functionality was improved during two iterations of the software. A moderately large database was populated for kinematics during the deadlift exercise, which informed the machine learning algorithms for the software. Using the VETSS, a randomized controlled trial was completed and data were analyzed. Data were lost from four participants at the time of this report, but these data will be salvaged prior to publication. Three additional exercises (dumbbell deadlift, goblet squat, and Romanian deadlift) were added to the database and the software tool was piloted during the NASA Extreme Environment Mission Operations (NEEMO) 22 mission. All tasks related to testing the virtual exercise software for treadmill running are complete. The acute and training response data were collected and are 100% analyzed. The specific aims were completely addressed but to better address exercise adherence resulting from the software, a longer duration study may be needed.
Bibliography Type:	Description: (Last Updated:)